

Database Setup & Planning Guide

AI Career Companion & Placement Acceleration Platform

Scale: **10,000+ Students**
Alumni Mentors: **3,000+ Placed**
Target Stack: **PostgreSQL (Supabase)**
Doc Version: **1.0 (Production)**

TOTAL STUDENT USERS 10,000+ Active campus enrollment	ALUMNI MENTORS 3,000+ Top Tier IT & Core placement	ESTIMATED DB SIZE < 150 MB Less than 2% of 8 GB Plan	COST PER STUDENT ₹0.21 / mo Approx. 21 paise / student
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1. Comparative Evaluation: Why Other Options Fail at 13,000 Users

When operating a campus placement platform supporting 10,000 students and 3,000 alumni simultaneously taking mock tests and practicing questions, storage architecture must guarantee **sub-10ms response times, high concurrency, and zero data corruption**.

Storage Option	Verdict for 13,000 Users	Concurrency & Speed	Security & Reliability
Google Sheets API	FATAL / UNUSABLE Hard quota of 60 req/min/user. Crashes when 20+ students click test buttons.	High latency (500ms–1.5s per row). Hits quota 429 lockouts.	Plain-text passwords & 2FA keys visible to anyone with link.
SQLite (File-based)	DEVELOPMENT ONLY Single-writer lock causes database is locked crashes during exams.	Fast reads (<2ms), but single-threaded write bottleneck.	Single file on disk. High risk of corruption if server restarts.
Self-Hosted MySQL	VIABLE ON VPS Requires manual Linux server maintenance, manual backups, and phpMyAdmin.	High concurrency (handles 1,000+ queries/sec).	Strong relational security, but requires regular manual patching.
Supabase (PostgreSQL)	STRONGLY RECOMMENDED Full enterprise PostgreSQL with an interactive spreadsheet UI for non-coders.	Ultra-high concurrency via built-in Supavisor pooling.	Enterprise-grade SSL/TLS, bcrypt hashing, automated daily backups.

2. Capacity, Storage & Bandwidth Sizing Calculations

A mathematical proof demonstrating exact resource utilization for 10,000 students and 3,000 alumni mentors:

Data Asset	Record Count	Avg. Record Size	Total Calculated Size	Included Capacity (Pro Tier)
Student Profiles & Login Credentials	10,000 rows	1.5 KB / student	~15.0 MB	8.0 GB (8,192 MB) (Database utilizes only ~1.8% of available storage)
Alumni Mentors & Interview Journeys	3,000 rows	2.8 KB / story	~8.4 MB	

Data Asset	Record Count	Avg. Record Size	Total Calculated Size	Included Capacity (Pro Tier)
Practice MCQs & Technical Questions	5,000 questions	2.0 KB / question	~10.0 MB	
Mock Test Submissions & Logs	150,000 attempts	0.6 KB / attempt	~90.0 MB	
Resume Uploads (PDF Documents)	10,000 PDF files	350 KB / PDF	~3.5 GB	100 GB Included Storage

Key Sizing Takeaway

Even after a full academic year with 10,000 registered students, 3,000 alumni mentors, and over 150,000 completed placement test sessions, the entire database consumes less than **150 MB** of text data and **3.5 GB** of resume PDF files—well below the 8 GB database and 100 GB file storage included in the base Pro tier!

3. Transparent Pricing & Cost Breakdown

Comparison between Supabase tiers and traditional enterprise cloud databases (AWS RDS / GCP Cloud SQL):

Feature / Resource	Free Tier (\$0 / month)	Pro Plan (\$25 / month ≈ ₹2,100)	AWS RDS PostgreSQL (Equivalent)
Included Database Storage	500 MB (Shared Compute)	8 GB Included (\$0.125 / extra GB)	20 GB (\$0.115 / GB + \$35 compute)
Monthly Active Users (MAU)	50,000 MAU	100,000 MAU Included	Unlimited (Limited by CPU/RAM)
Resume Document Storage	1 GB (holds ~3,000 resumes)	100 GB Included (~250,000 resumes)	Separate AWS S3 billing required
Monthly Network Bandwidth	5 GB	250 GB Included	Data transfer egress charges apply
Automated Database Backups	Manual dump only	Daily Automated Backups (7-day retention)	Automated snapshot fees apply
Connection Pooler (Supavisor)	Shared micro pool (~200 conns)	Dedicated High-Concurrency Pooling	Requires AWS RDS Proxy (~\$15/mo extra)
Project Pausing Policy	Pauses after 7 days inactivity	Never Pauses (24/7/365 Uptime)	Never pauses (Billed hourly)
Total Monthly Cost	₹0 (Free)	≈ ₹2,100 / month flat	≈ ₹6,500 – ₹12,000 / month

4. Non-Technical / Zero-Code Data Management Workflow

College placement cells, faculty advisors, and departmental heads do not need to write SQL or run commands. They manage data across two non-technical layers:

Layer 1: Built-in Admin Portal with 1-Click Excel Batch Sync (Daily Use)

Placement coordinators manage students and placement records directly inside the website's Admin Portal using familiar Excel workflows:

- **Batch Student Onboarding:** Download the pre-formatted Excel template, paste 10,000 roll numbers, names, emails, and branches from the college ERP/admissions desk, and click "*Upload Excel/CSV*". The system provisions accounts and 2FA credentials in seconds.
- **Audit & Accreditation Export:** Click "*Export Placement Report (.xlsx)*" to generate formatted reports ready for NAAC, NBA, and NIRF inspection visits with single-click ease.

Layer 2: Supabase Visual Spreadsheet Table Editor (Direct Visual Edits)

Non-coders access the secure cloud web dashboard at supabase.com in any browser:

- **Spreadsheet Grid Interface:** Looks, feels, and edits like Google Sheets or Airtable. Double-click any student cell to update phone numbers, placement status, or offered packages.
- **Visual Filtering & Search:** Filter students by Branch (e.g. CSE), Placement Status (e.g. Placed), or Company (e.g. Amazon) with standard dropdown menus.

- **Role-Based Access:** Assign read-only viewing to junior faculty while reserving master write permissions for the Head of Training & Placement (TPO).

5. Step-by-Step Technical Setup Blueprint

1. **Sign Up on Supabase:** Visit <https://supabase.com> → Click "New Project" → Set Project Name: ai-career-companion → Region: ap-south-1 (Mumbai, India) for lowest student latency (< 25ms across Indian campuses).
2. **Copy Database Credentials:** In Project Settings → Database → Connection String → Copy the URI format for Connection Pooling (Port 6543).
3. **Configure Environment Variable:** In your server's .env file, set:

```
DATABASE_URL = "postgresql://postgres.xxx:[PASSWORD]@aws-0-ap-south-1.pooler.supabase.com:6543/postgres?sslmode=require"
```

4. **Execute Schema Migration:** Run the one-time migration script (schema shown below) in Supabase SQL Editor.
5. **Switch Connection Driver:** Update database.py to connect using SQLAlchemy or psycopg2-binary with connection pooling enabled.

6. PostgreSQL Production Database Schema Definition

```
-- ===== -- 1. STUDENTS & PLACEMENT READINESS TABLE
(10,000+ Students) -- ===== CREATE TABLE IF NOT EXISTS
students ( id SERIAL PRIMARY KEY, name VARCHAR(120) NOT NULL, roll_no VARCHAR(50) UNIQUE NOT NULL, phone
VARCHAR(25) DEFAULT '', email VARCHAR(120) NOT NULL, branch VARCHAR(80) NOT NULL, section VARCHAR(10)
DEFAULT 'A', password_hash VARCHAR(255) NOT NULL, batch_year VARCHAR(10) NOT NULL, ats_score NUMERIC(5,2)
DEFAULT 0.00, readiness_status VARCHAR(50) DEFAULT 'In Progress', target_company VARCHAR(100) DEFAULT
'General', quizzes_completed INTEGER DEFAULT 0, placement_status VARCHAR(50) DEFAULT 'Not Placed',
placed_company VARCHAR(100) DEFAULT '', package_lpa NUMERIC(6,2) DEFAULT 0.00, created_at TIMESTAMPTZ
DEFAULT NOW() ); CREATE INDEX IF NOT EXISTS idx_students_roll_no ON students(roll_no); CREATE INDEX IF NOT
EXISTS idx_students_placement ON students(placement_status, branch); --
===== -- 2. ALUMNI MENTOR ACCOUNTS TABLE (3,000+ Placed
Seniors) -- ===== CREATE TABLE IF NOT EXISTS
alumni_accounts ( id SERIAL PRIMARY KEY, name VARCHAR(120) NOT NULL, alumni_id VARCHAR(50) UNIQUE NOT NULL,
email VARCHAR(120) NOT NULL, phone VARCHAR(25) DEFAULT '', branch VARCHAR(80) NOT NULL, batch_year
VARCHAR(10) NOT NULL, company VARCHAR(100) NOT NULL, role VARCHAR(120) NOT NULL, package_lpa NUMERIC(6,2)
DEFAULT 0.00, password_hash VARCHAR(255) NOT NULL, security_key_2fa VARCHAR(100) NOT NULL, created_at
TIMESTAMPTZ DEFAULT NOW() ); -- ===== -- 3. ALUMNI
INTERVIEW EXPERIENCES & ROUNDS TABLE -- ===== CREATE
TABLE IF NOT EXISTS alumni_experiences ( id SERIAL PRIMARY KEY, student_name VARCHAR(120) NOT NULL,
batch_year VARCHAR(10) NOT NULL, email VARCHAR(120) DEFAULT 'alumni@college.edu', company VARCHAR(100) NOT
NULL, role VARCHAR(120) NOT NULL, package_lpa NUMERIC(6,2) NOT NULL, offer_type VARCHAR(50) DEFAULT 'On-
Campus', difficulty VARCHAR(30) DEFAULT 'Medium', status VARCHAR(30) DEFAULT 'Selected', rounds_json JSONB
NOT NULL, preparation_tips TEXT NOT NULL, advice_to_juniors TEXT NOT NULL, upvotes INTEGER DEFAULT 0,
created_at TIMESTAMPTZ DEFAULT NOW() ); -- ===== -- 4.
PRACTICE MCQs & ALUMNI CONTRIBUTIONS TABLE -- =====
CREATE TABLE IF NOT EXISTS mcq_questions ( id SERIAL PRIMARY KEY, subject VARCHAR(80) NOT NULL, topic
VARCHAR(100) NOT NULL, company VARCHAR(100) DEFAULT 'General', question TEXT NOT NULL, option_a TEXT NOT
NULL, option_b TEXT NOT NULL, option_c TEXT NOT NULL, option_d TEXT NOT NULL, correct_option CHAR(1) NOT
NULL, explanation TEXT NOT NULL, difficulty VARCHAR(30) DEFAULT 'Medium', alumni_id VARCHAR(50) DEFAULT
NULL, alumni_name VARCHAR(120) DEFAULT 'Alumni Mentor', created_at TIMESTAMPTZ DEFAULT NOW() ); CREATE INDEX
IF NOT EXISTS idx_mcqs_filters ON mcq_questions(subject, company, difficulty, alumni_name);
```

7. Enterprise Security Architecture

- **Password Hashing (bcrypt):** Passwords must never be stored in plain text. Use Python's `werkzeug.security.generate_password_hash` with salt rounds ≥ 12 .
- **Two-Factor Authentication (2FA):** Roll numbers and Alumni IDs act as unique security seeds, verified in separate authorization stages before session tokens are issued.
- **Connection Encryption (TLS 1.3):** All database traffic between your Flask web server and Supabase PostgreSQL is encrypted using modern SSL certificates.
- **Disaster Recovery & Point-in-Time Recovery:** The Pro plan takes daily physical snapshots and retains Write-Ahead Logs (WAL), enabling point-in-time recovery down to the exact second.